TASK 4: VERSION-CONTROLLED DEVOPS PROJECT WITH GIT

Solution and GitHub Submission Guide

1. Objective

Manage a DevOps project using Git best practices with Git and GitHub. Deliverables include a repository with meaningful commits and branches, pull requests, a README, .gitignore, tags, and Markdown documentation.

2. Repository Structure

• **README.md** — project overview and quick start

• **.gitignore** — ignores temporary files, build output, and local secrets

• **CHANGELOG.md** — initial release notes

• **docs/git-workflow.md** — complete workflow and commands

• **docs/interview-questions.md** — answers to all eight interview questions

• **scripts/git-workflow-demo.sh** — optional local branch demonstration

3. Branching Strategy

main: stable, reviewed version. **dev:** integration branch for ongoing work. **feature/<name>:** focused work branch created from dev and merged through a pull request.

4. Publish the Project to GitHub

1. Create a GitHub repository named task4-version-controlled-devops. Do not initialize the remote with a separate README if you are pushing this existing project.

2. In the project directory, run: git init; git branch -M main; git add .; git commit -m "chore: initialize Task 4 DevOps Git project".

3. Add the remote: git remote add origin https://github.com/YOUR_USERNAME/task4-version-controlled-devops.git

4. Push main: git push -u origin main

5. Create and push dev: git checkout -b dev; git push -u origin dev

6. Create and push a feature branch: git checkout dev; git checkout -b feature/add-git-documentation; git push -u origin feature/add-git-documentation

7. On GitHub, create a PR from feature/add-git-documentation into dev. Review and merge it. Then create a PR from dev into main and merge after review.

8. Tag a release on main: git checkout main; git pull origin main; git tag -a v1.0.0 -m "Task 4 initial release"; git push origin v1.0.0

Replace YOUR_USERNAME with your GitHub username. If origin already exists, use git remote -v and git remote set-url origin <URL> rather than adding it again.

5. Verification Commands

```
git status
git branch --all
git log --oneline --graph --decorate --all
git remote -v
git tag
```

6. Interview Questions and Answers

What is Git?

A distributed version-control system that records file changes and supports collaboration and recovery.

Merge vs rebase?

Merge combines histories and often creates a merge commit. Rebase replays commits onto a new base for a more linear history; avoid rebasing shared commits without team agreement.

What is a pull request?

A GitHub request to review changes and merge one branch into another, with discussion and checks before merging.

How do you resolve merge conflicts?

Check git status, edit conflicted files and remove markers, test, git add the resolved files, then complete the merge commit. Use git merge --abort to cancel an unfinished merge.

What are Git tags?

Names attached to specific commits, commonly used for releases such as v1.0.0. Annotated tags include a message and metadata.

What is Git workflow?

The agreed process for branching, committing, reviewing, merging, and releasing. Here, main is stable, dev integrates work, and feature branches hold focused changes.

Explain git stash.

Temporarily saves uncommitted changes so you can switch context. Use git stash list to inspect and git stash pop to reapply the latest stash.

Why use .gitignore?

It excludes specified untracked files such as logs, build output, dependency folders, and local environment files. It does not untrack files already committed.

7. Submission Checklist

- ■ GitHub repository created and URL copied
- ■ main, dev, and feature/* branches pushed
- ■ Feature PR merged into dev
- ■ dev PR merged into main
- ■ .gitignore and README.md included
- ■ Markdown task documentation included
- ■ v1.0.0 tag pushed
- ■ Branches, commits, remotes, and tags verified

Important: The files and instructions are prepared. Actual creation of the GitHub repository, authentication, pushing, and PR merges must be performed in your GitHub account.